

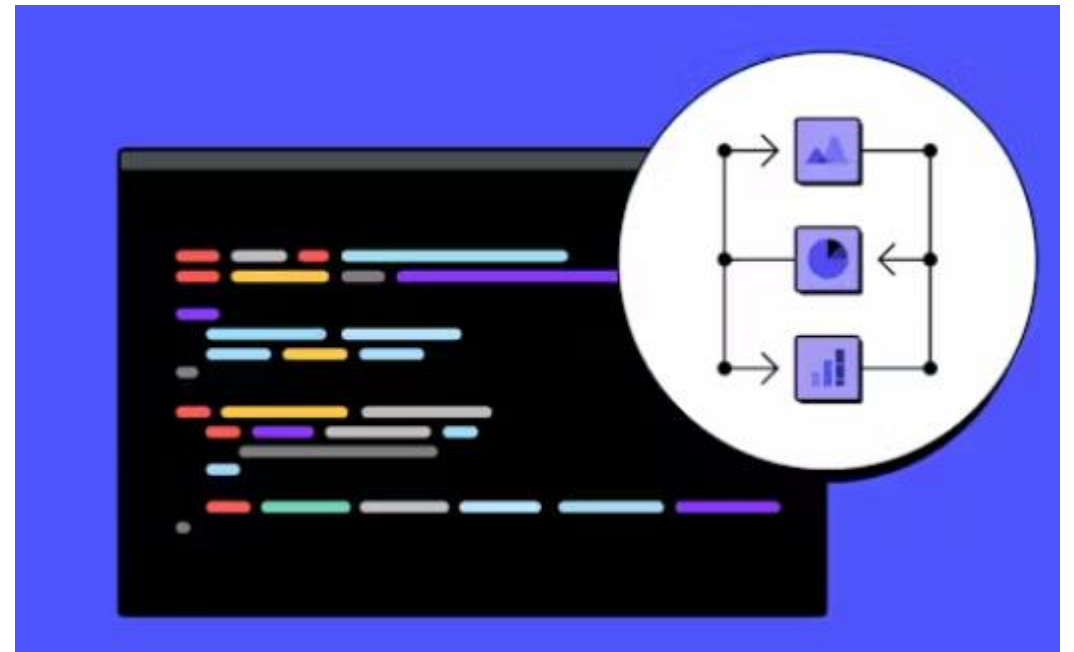
Data Modeling



Business Intelligence

LEARNING OUTCOMES

Students are able to explain the concept of data modelling in a data warehouse



TOPIC



1. Architecture
2. Introduction to modeling
3. Schema Datawarehouse



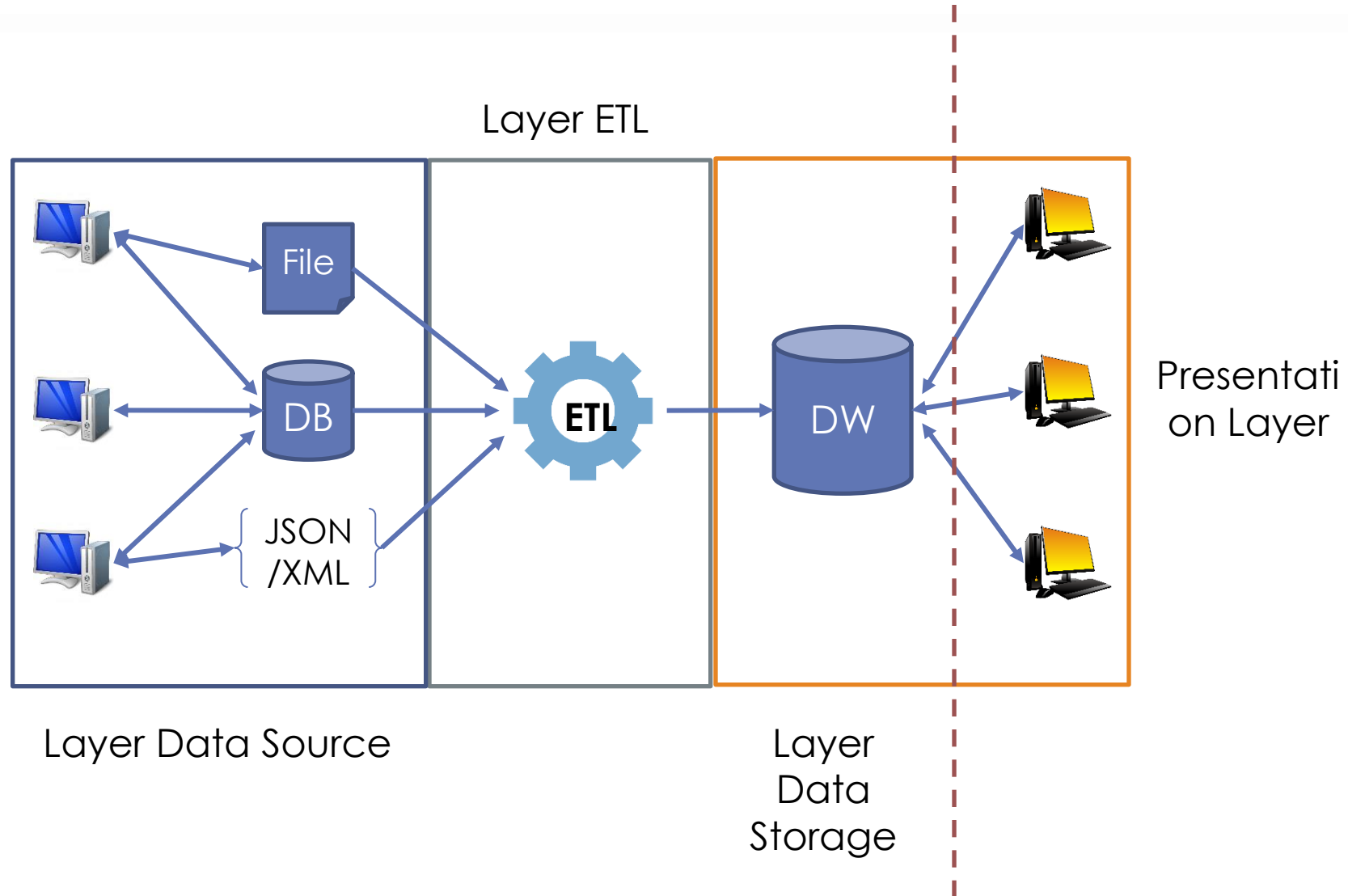
ARCHITECTURE

ARCHITECTURE SUMMARY

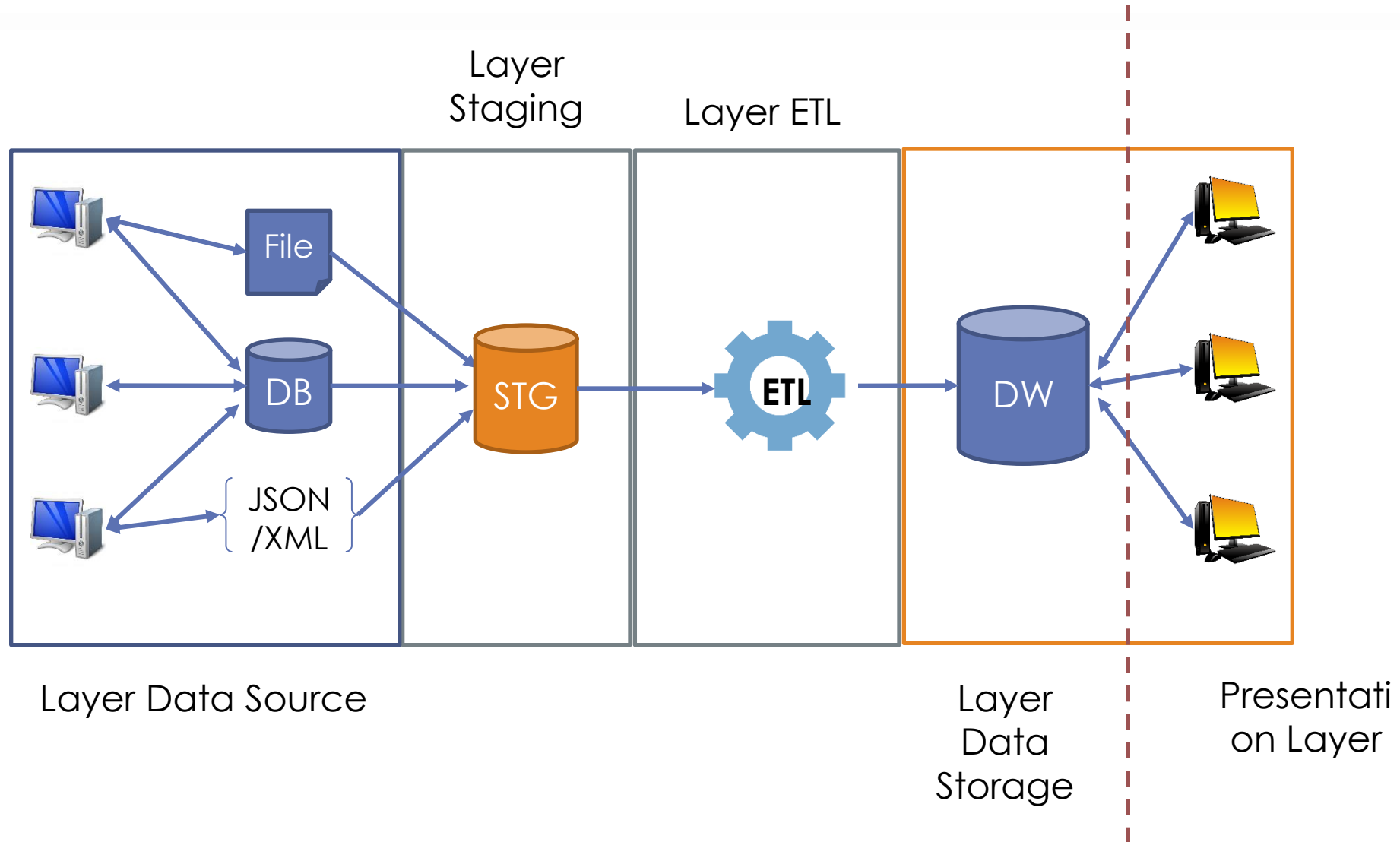


- Data Warehouse can be implemented in a wide variety of device configurations.
- Adapt to the needs of the company
- At a minimum configuration, there are generally 3 components
- **Layer-1: OLTP + Data Source**
- **Layer-2: ETL**
- **Layer-3: Data Warehouse + OLAP**
- **In other configurations, usually between Layer-1 and Layer-2 Staging is added**
- Various data sources are collected first into 1 database.
- Components on each layer can be placed on 1 or more devices.

ARCHITECTURE MINIMAL CONFIGURATION



ARCHITECTURE CONFIGURATION WITH STAGING



ARCHITECTURE STAGING



- Data from various sources is retrieved, cleaned, tidied, and collected first before being exposed to be retrieved by ETL.
- Advantage: ETL's work is lighter because there is no need to do data cleansing.
- Deficiency: Architecture has become more complicated.

ARCHITECTURE TYPES



- According to the place where the ETL engine is placed consists of 2 types:
- **Shared** Architecture
- **Dedicated** Architecture

Shared Architecture

- ETL is placed in 1 machine along with data source and/or data storage.

Dedicated Architecture

- ETL is placed in 1 machine separately from the data source and data storage
- Which one to choose?
- A little money -> Shared
- A lot of money -> Dedicated



INTRODUCTION TO MODELING



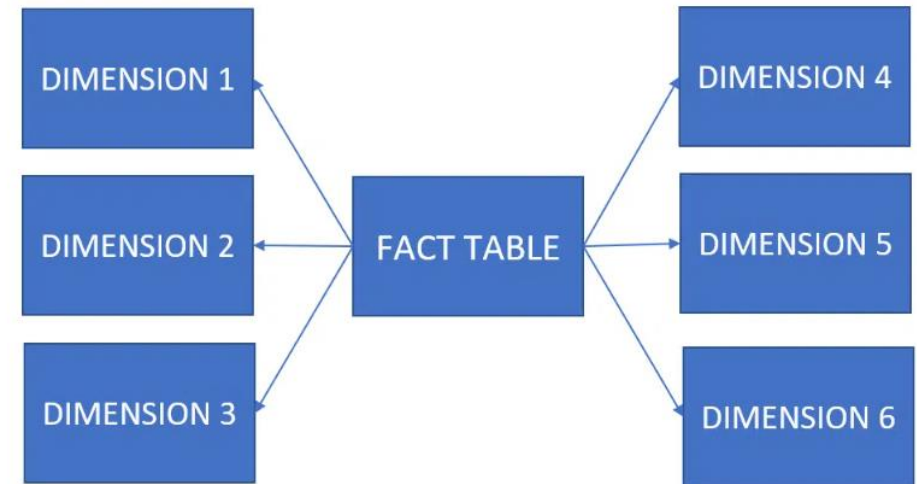
DATA WAREHOUSE MODELING

- Data warehouse modeling is the process of designing and organizing your data models within your data warehouse platform.
- The design and organization process consists of setting up the appropriate databases and schemas so that the data can be transformed and then stored in a way that makes sense to the end user.
- To succeed, it is key to design your data warehouse with your models in mind before starting the modeling process.

DIMENSIONAL MODELING

Dimensional modeling is a data modeling technique used in data warehousing that allows businesses to structure data to optimize analysis and reporting. **This method involves organizing data into dimensions and facts, where dimensions are used to describe the data, and facts are used to quantify the data.**

For instance, suppose a business wants to analyze sales data. In that case, the dimensions could include customers, products, regions, and time, while the facts could be the number of products sold, the total revenue generated, and the profit earned.





SCHEMA





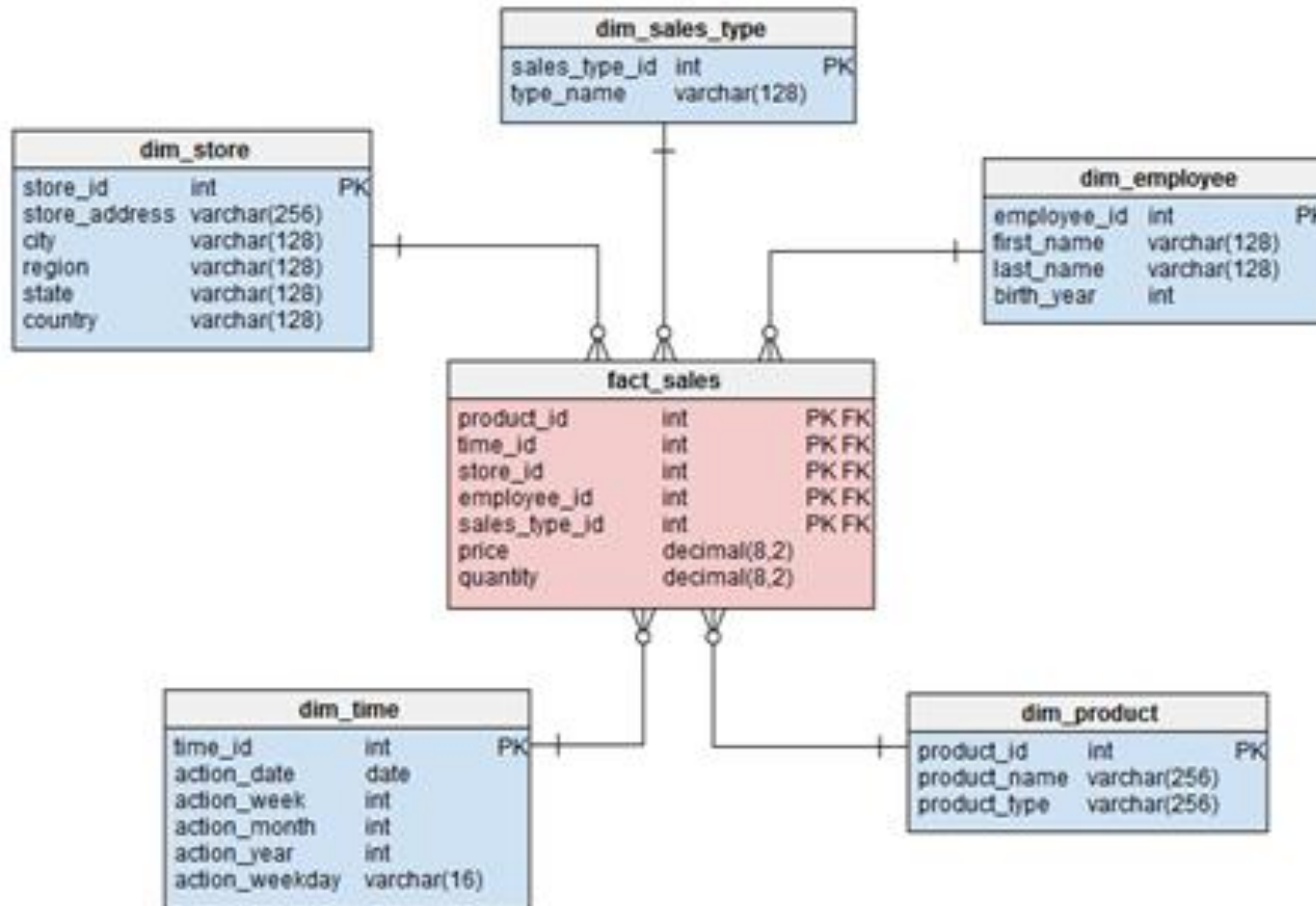
- Remember! The most important differences between a regular database in OLTP and a data warehouse in OLAP are: THE DESIGN/SCHEME
- **OLTP Common Database → Normalization**
 - Optimized for WRITE operations.
- **Database Data Warehouse → Denormalization**
 - Optimized for Read Operations
- There are 3 known Data Warehouse Schemes today
 - Star Schema
 - Snowflake Schema
 - Star Constellation Schema

Tabel Dimensi dan Tabel fakta

- A dimension table is a table that stores information about the dimensions or characteristics of the data measured in a data warehouse.
- Dimensions are different angles of data, such as time, location, product, or customer.
- Examples of attributes in a time dimension table can include date, month, year, day, hour, etc.
- Dimension tables typically have a relatively flat structure and consist of a primary key and several descriptive attributes.

- A fact table is a table that stores the data that is measured or observed in a data warehouse, as well as numerical information related to a specific dimension.
- Facts in a fact table are often the size or metrics of a business, such as sales, revenue, or the number of items sold.
- Each row in the fact table represents an event or business transaction that is observed and recorded.
- A fact table usually consists of two types of columns: dimension keys that serve as foreign keys to the dimension table and columns that store the factual values.

STAR SCHEMA

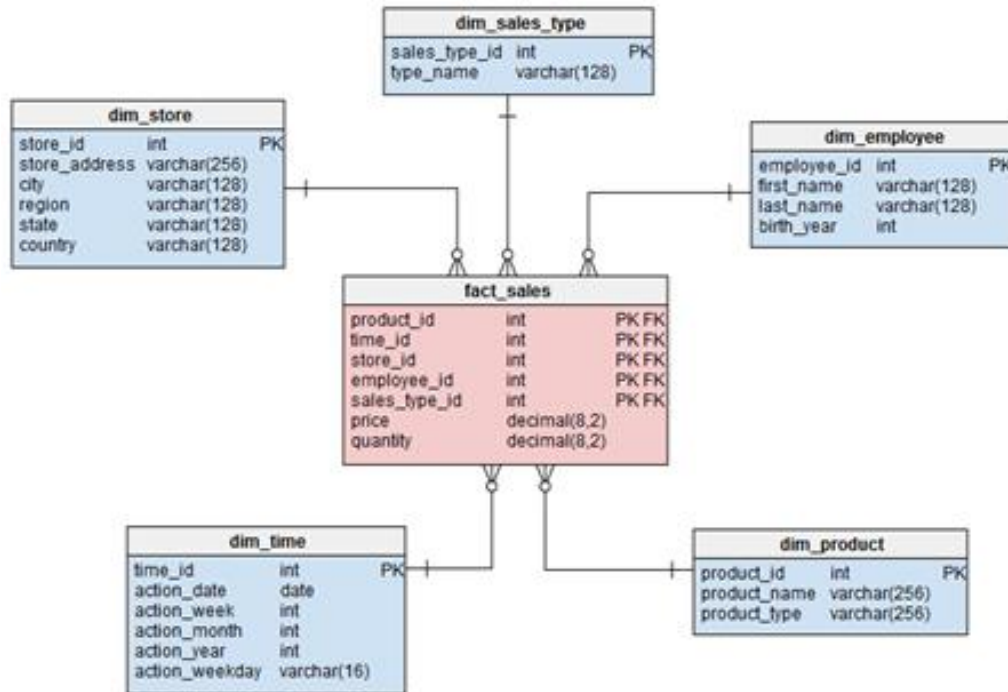


Simple and Easy to Understand:

Star schema has a simple and easy-to-understand structure, making it one of the most popular models in data warehouse design.

The clear and intuitive structure facilitates fast data capture and analysis.

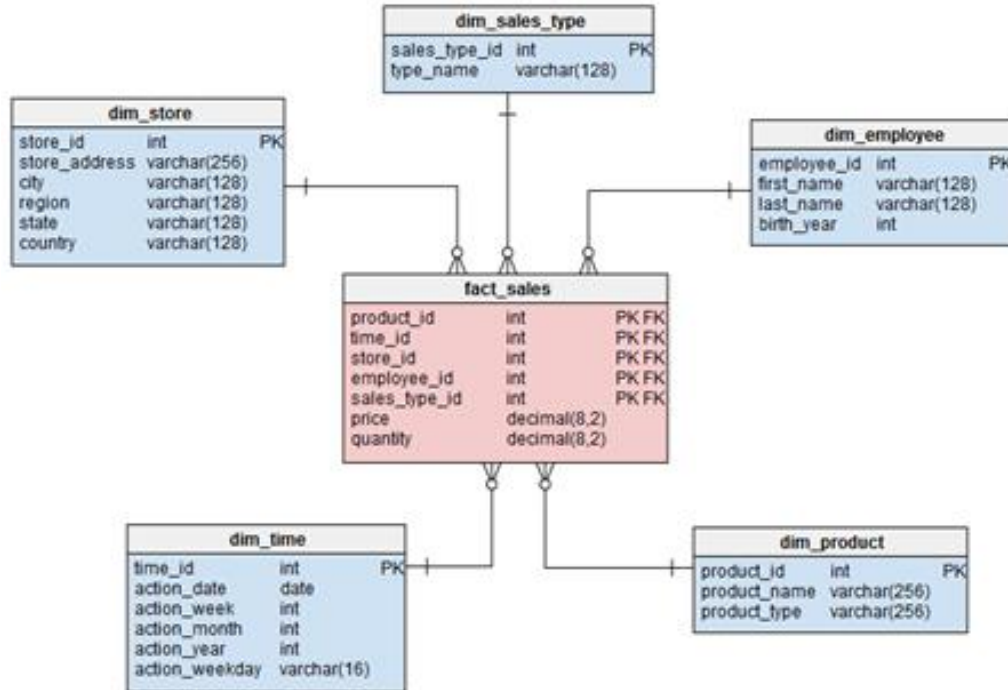
STAR SCHEMA



1.Fact Table:

1. The fact table serves as the central point in the star schema.
2. Fact tables store measured or observed data, such as business metrics or sales figures.
3. Each row in the fact table represents an observed business event or transaction.
4. Fact tables usually have dimension keys (foreign keys) that connect them to the dimension table.

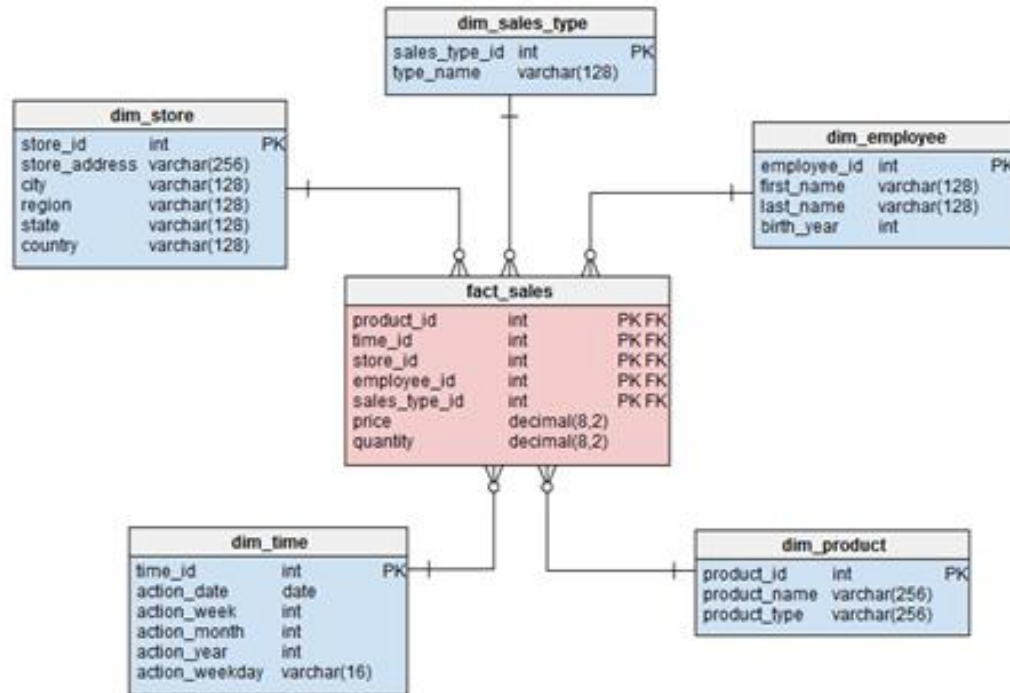
STAR SCHEMA



2.Dimension Table:

1. Dimension tables store information about the dimensions or characteristics of the data measured in the data warehouse.
2. Dimensions can be time, product, customer, or any other dimension relevant to the analysis.
3. Each dimension table typically has one row for each unique value in that dimension.
4. A dimension table has a one-to-many relationship with a fact table, where a single row in a dimension table can relate to multiple rows in a fact table.

STAR SCHEMA

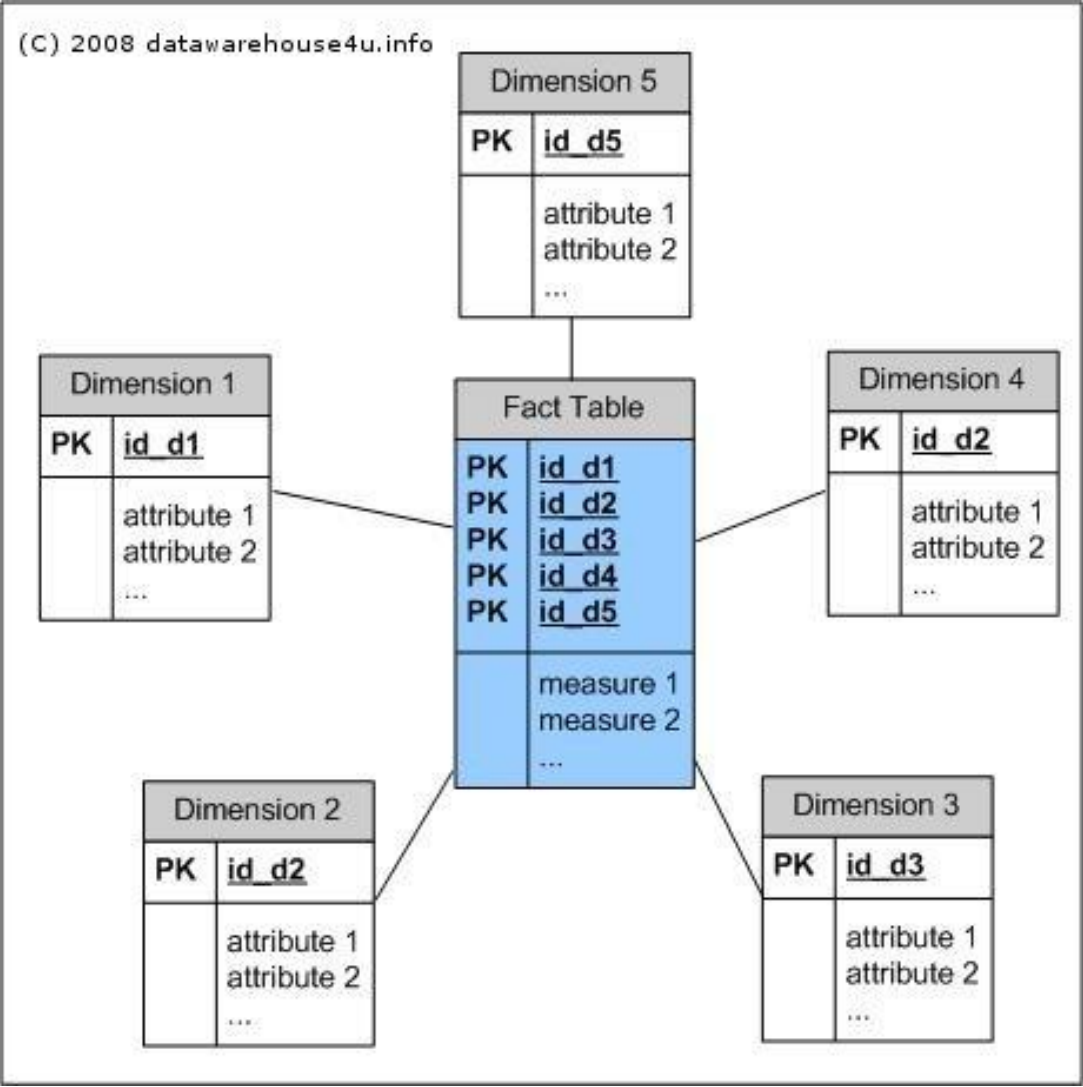


3. Relationships between Tables:

1. The relationship between a fact table and a dimension table is star, where the fact table serves as the center and the dimension table serves as the "arm" of the star that contacts it.
2. This relationship is usually done through dimension keys, where the dimension keys in the fact table are the foreign keys that connect them to the dimension table.

Star Schema

- Is the simplest Schema
- It is called so because it is like a star
- Most commonly used when in data warehouses
- Recommended by Oracle
- The one in the middle is called: Fact Table
- What do you want to analyze?
- The one on the edge is called: Dimension Table
- Analyzed based on what?
- 1 Fact tables can relate to 1 or more dimension tables.
- 1 star is also called 1 data mart

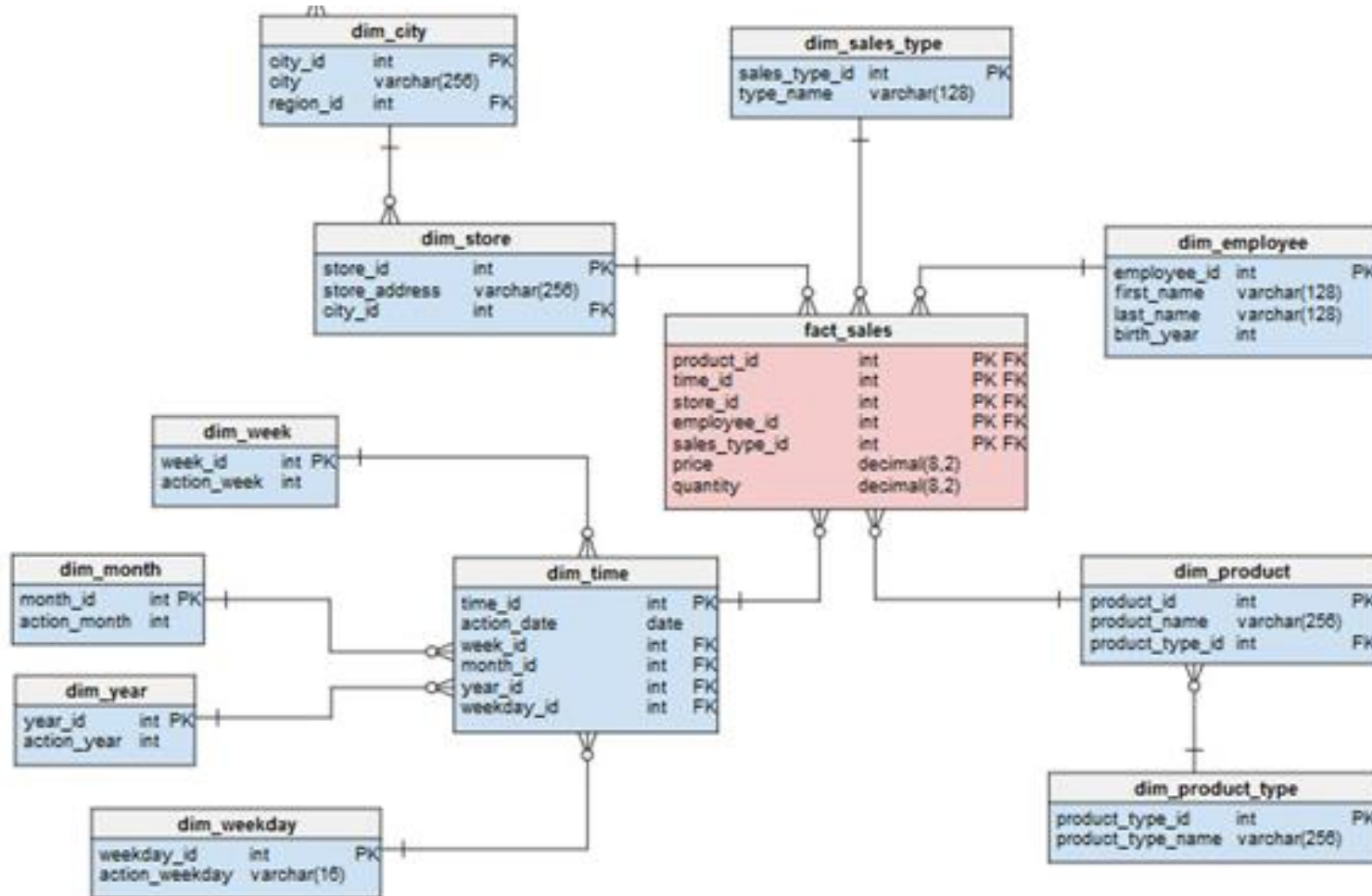


Schema

Star Schema Characteristics

- The structure is simple Easy to understand
- High query effectiveness
- Fewer JOIN tables
- The most common schema used in data warehouse implementation.
- Takes longer to populate data to dimension tables
- The process of denormalization
- Redundant data causes the size to be relatively larger

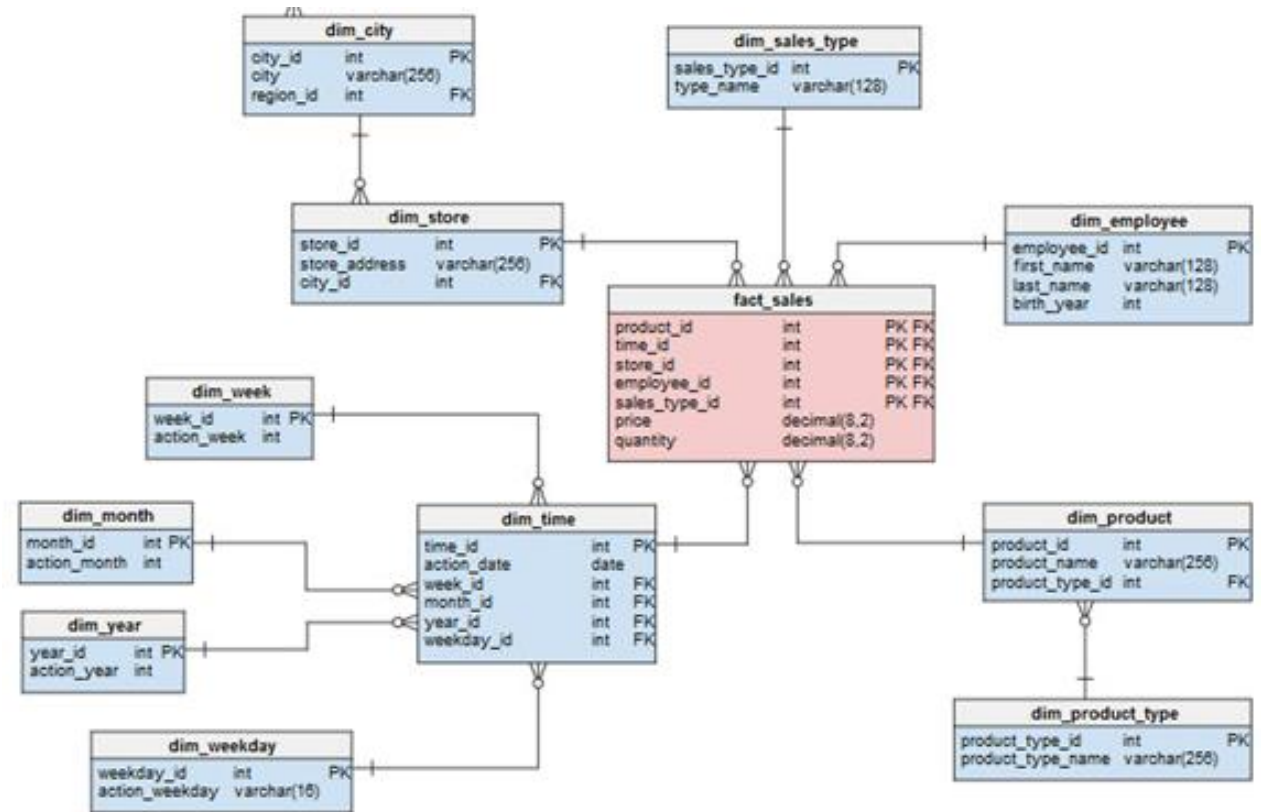
SNOWFLAKE SCHEMA



SNOWFLAKE SCHEMA

1. Fact Table:

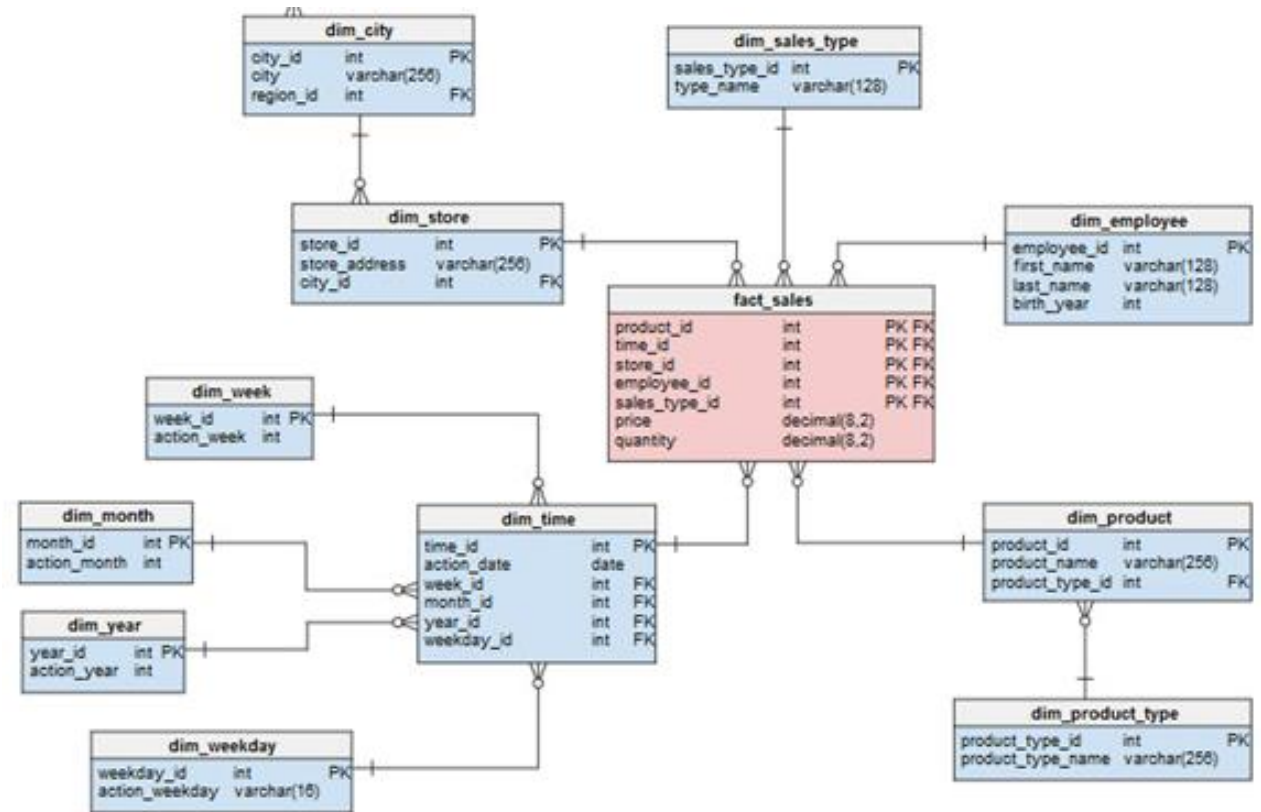
1. The fact table is the center of the snowflake schema, which stores the measured or observed data, as well as the associated dimension keys.
2. Each row in the fact table represents an event or business transaction that is observed and recorded.
3. The columns in the fact table include dimension keys that connect them to the dimension table, as well as fact attributes such as business metrics or size.



SNOWFLAKE SCHEMA

2. Dimension Table:

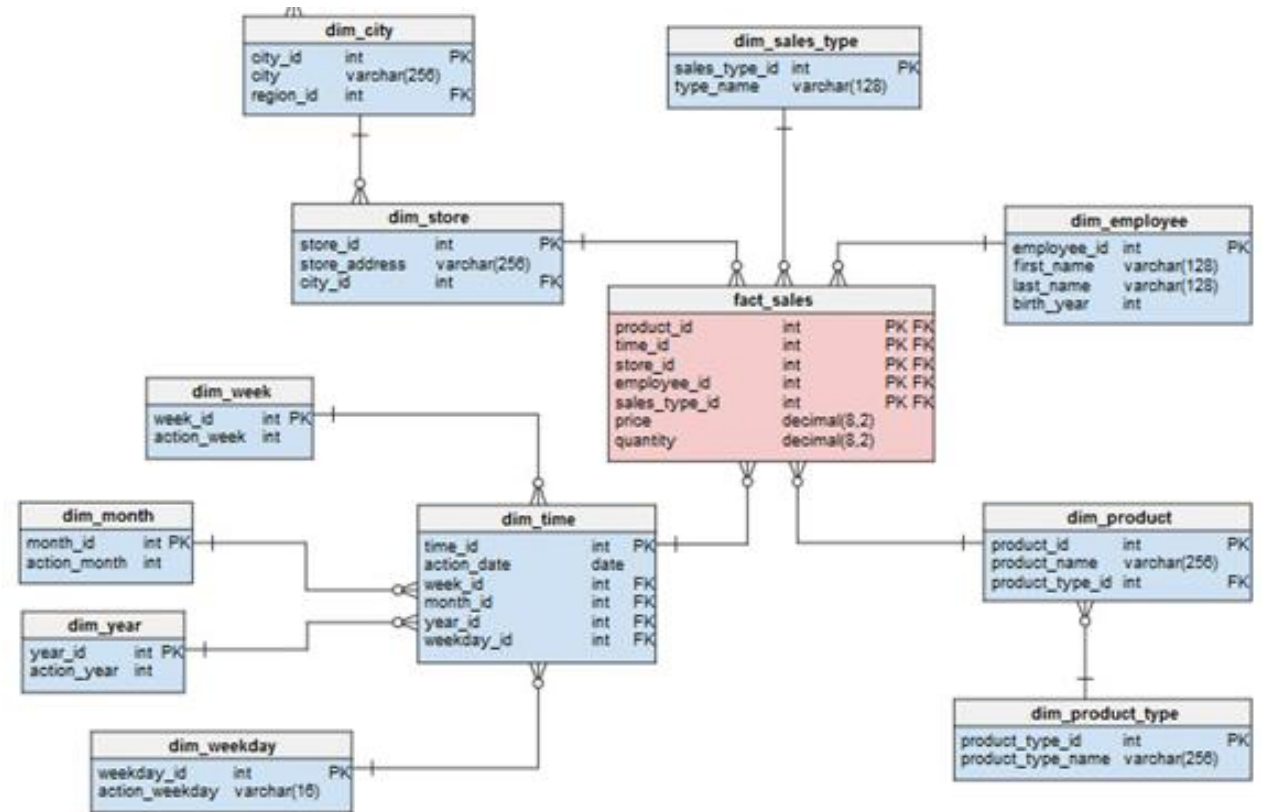
1. A dimension table is a table that stores information about the dimensions or characteristics of the data measured in a data warehouse.
2. These dimensions can include information such as time, product, customer, and location.
3. In a snowflake schema, a dimension table can be divided into smaller or normalized tables, forming "branches" or "branches" that lead to more detailed or specific dimensions.



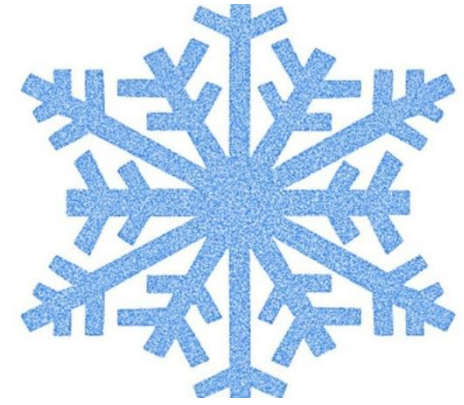
SNOWFLAKE SCHEMA

3. Relationships between Tables:

1. The relationships between tables in a snowflake schema illustrate how the fact tables are connected to the dimension table and how the dimension tables might be connected to each other.
2. The relationship between tables usually occurs through the foreign keys in the fact table that refer to the primary keys in the dimension table.
3. In addition, in a snowflake schema, the dimension tables can be connected to each other through foreign keys that form the normalization structure.

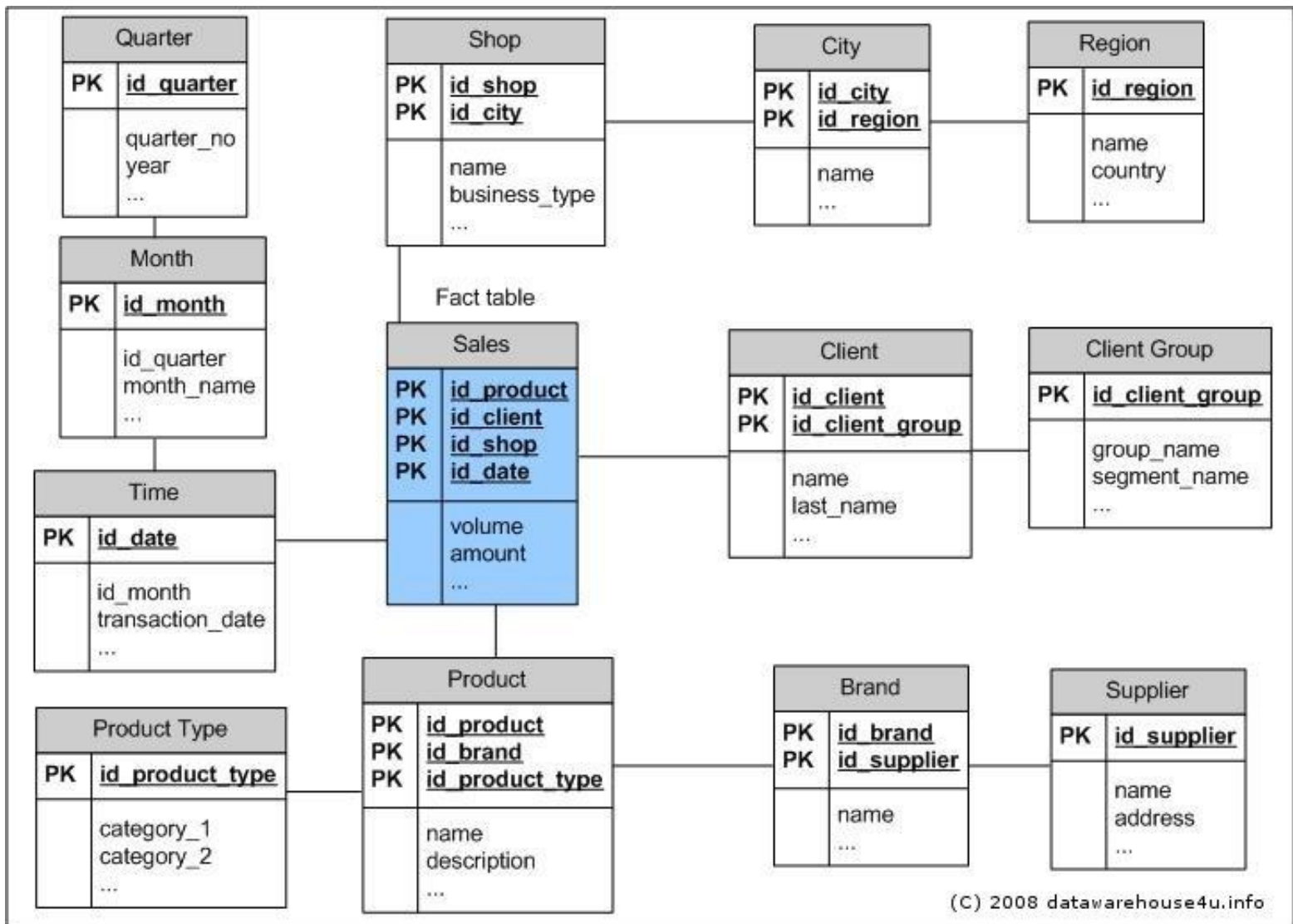


Snowflake Schema



- Snowflake
- More complex structures than Star-Schema
- Similar to star-schema, only each dimension table is normalized
- The dimensions are broken down, have a relationship
- Facilitate the drill-down or analysis process with a certain level of data specificity.
- Example:
- Sales based on Province, Province, District, District.... Dst.

Snowflake Schema



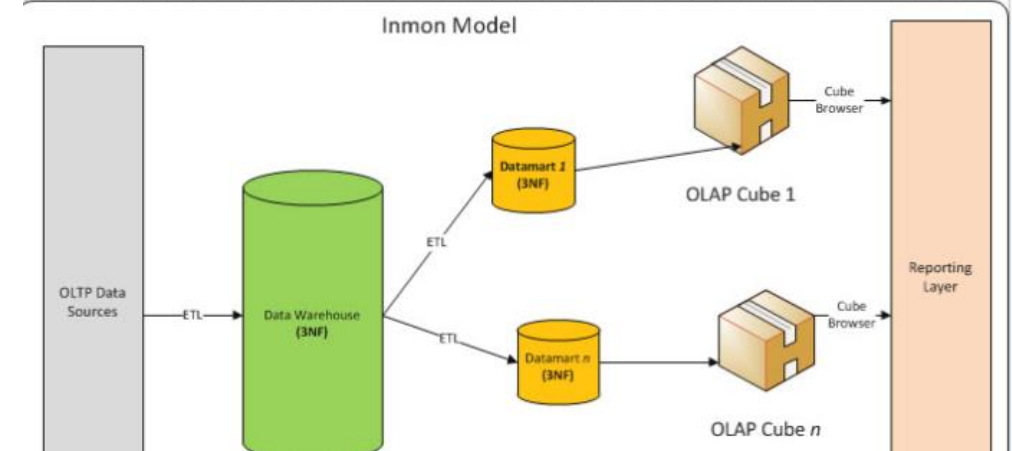
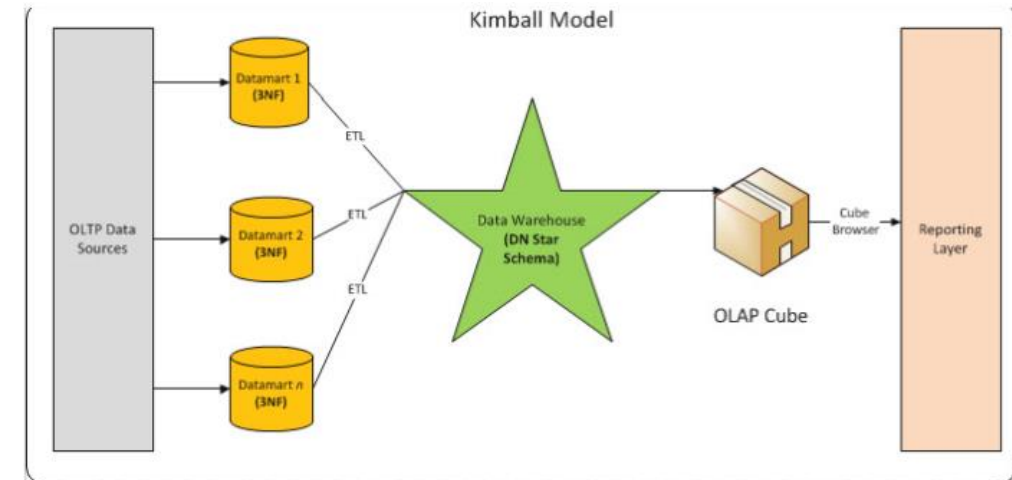
DIMENSIONAL MODELING TECHNIQUES

Kimball and **Inmon** are two popular methodologies

Ralph Kimball



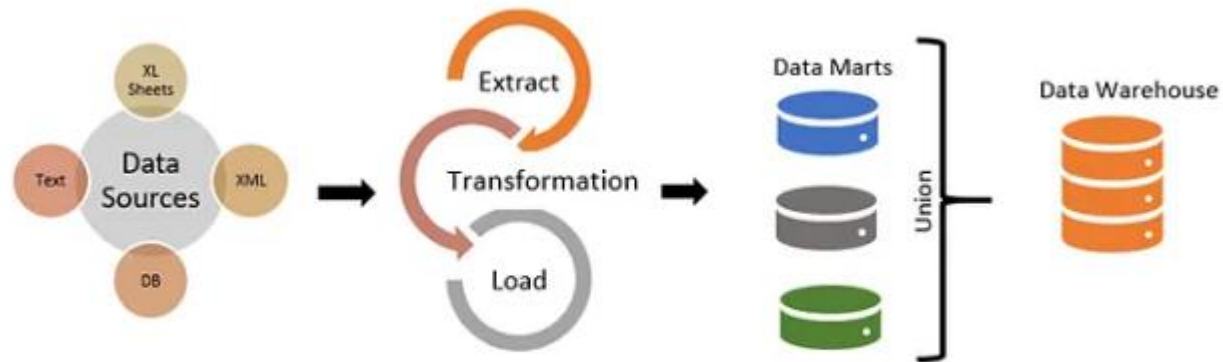
Bill Inmon



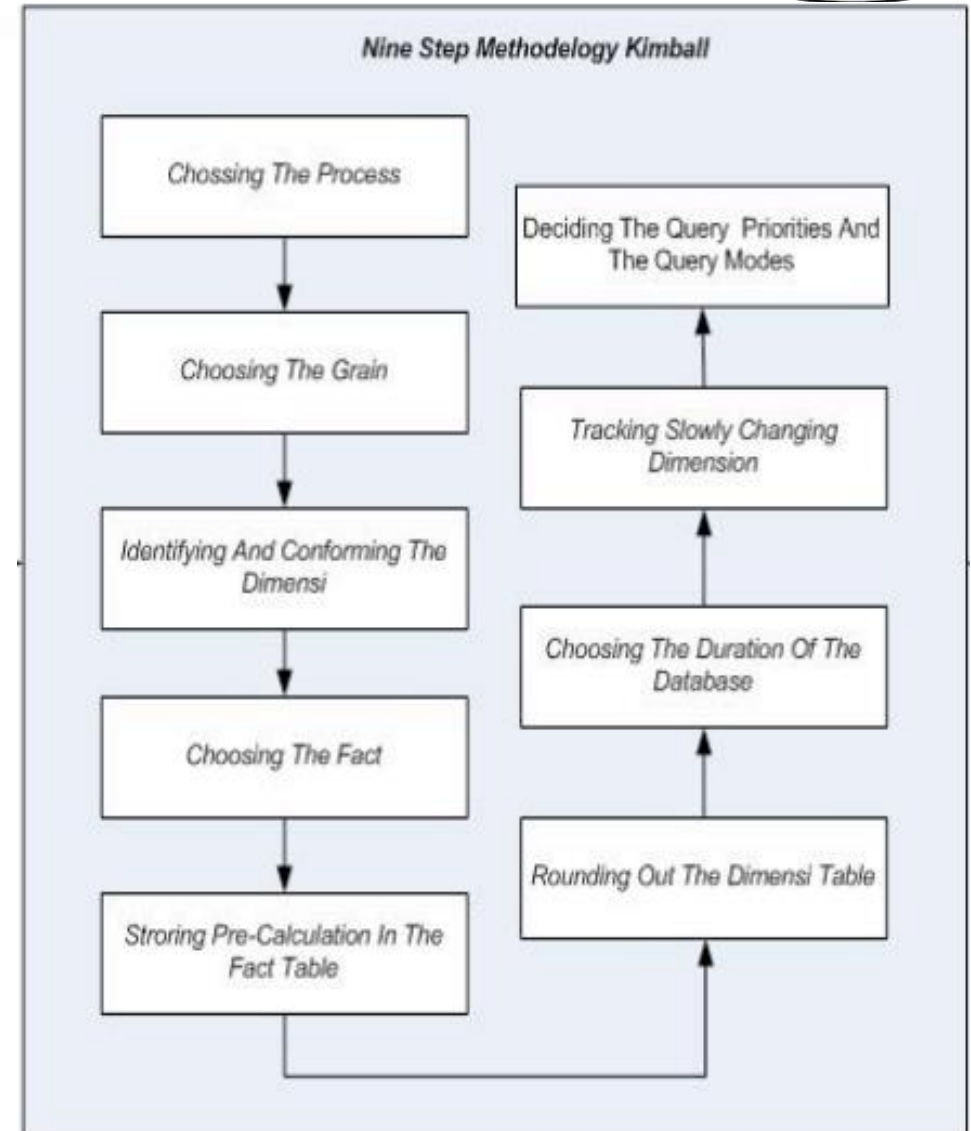
<https://medium.com/cloudzone/inmon-vs-kimball-the-great-data-warehousing-debate-78c57f0b5e0e>

TEKNIK DIMENSIONAL MODELING

- **nine step methodology Kimball :**



Diagrammatic representation of Kimball Approach



TEKNIK DIMENSIONAL MODELING

Inmon methodology

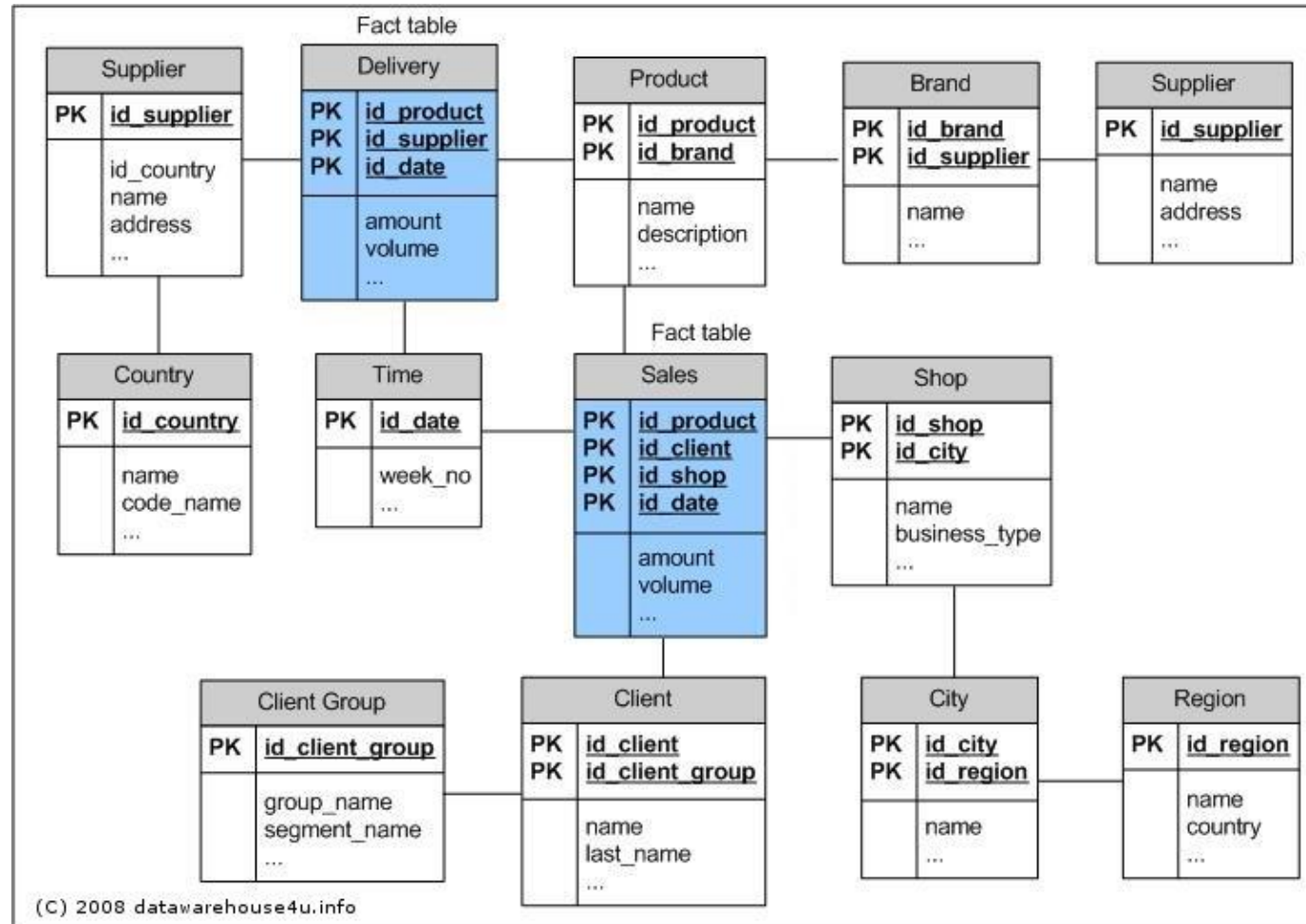


Diagrammatic representation of Inmon Approach

SKEMA

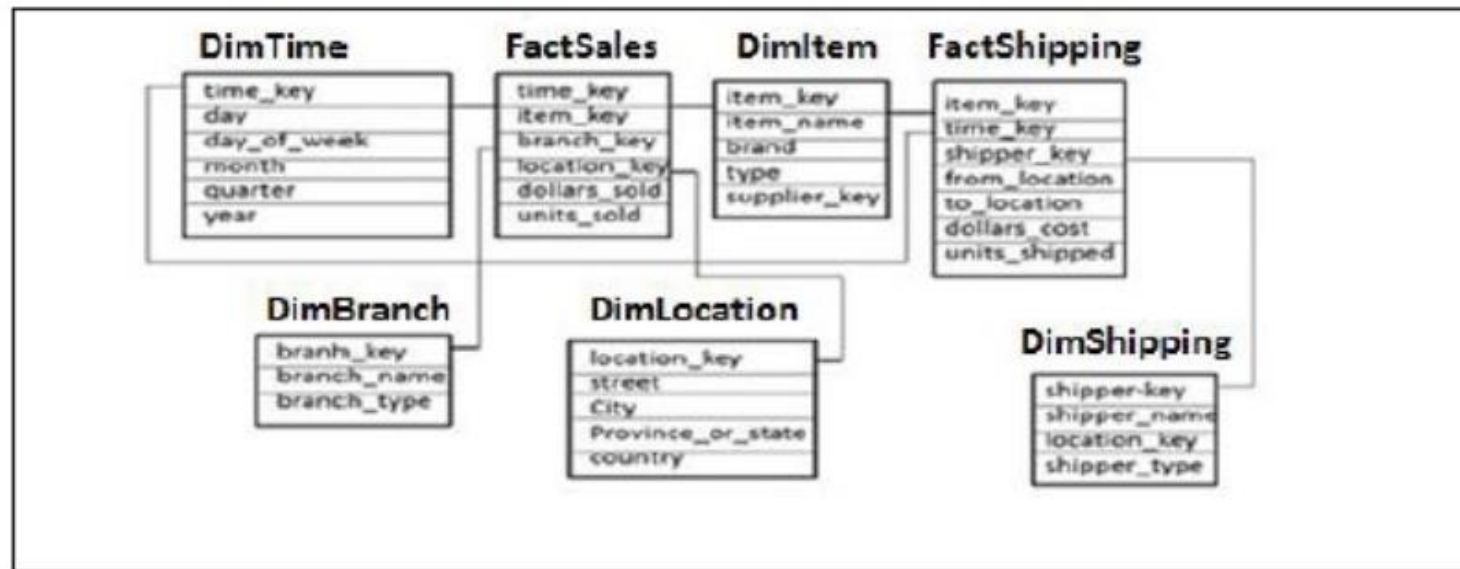
STAR-CONSTELLATION SCHEMA/GALAXY SCHEMA

- Star Schema yang masing-masing star-nya saling terhubung satu sama lain.



Galaxy Schema

In Galaxy Schema, there are multiple Fact tables and Dimension tables. Each Fact table stores primary keys of few Dimension tables and measures/facts to do analysis.



In the above example, there are two Fact tables FactSales, FactShipping and multiple Dimension tables joined to Fact tables. Each Fact table contains Primary Key for joined Dim tables and measures/Facts to perform analysis.

TASK 1 (CASE STUDY)



- go on to the next jobsheet



THANK YOU

